A Qubit Efficient Hybrid Quantum Classical Framework for Distributed Order Management

Pauli Correlation Encoding with Exact MILP Recourse

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Executive Summary

Distributed Order Management (DOM) decides whether an order stays at its default distribution centre or moves to an alternate one when inventory or capacity falls short. We built a hybrid solver in which a three-qubit Pauli Correlation Encoding (PCE) circuit proposes the keep or divert call for each order, and an exact mixed integer program works out everything downstream as fulfilled cases, pallets, shortfall penalties, freight, and resource feasibility.
On a contested eight-order batch spanning 95 SKUs, the hybrid method lands on the same assignment as exhaustive enumeration of all 256 possibilities: objective value **$1,080,634.47**, zero optimality gap, using three qubits instead of the eight a direct encoding would need a **62.5% reduction** in register size. A 20-trial noisy-hardware simulation shows every misread decision landing exactly one bit away from the true optimum, and a cheap classical repair step recovers the exact answer in all 20 trials.

Metric	Default plan	Optimized plan
Objective value	$1,047,384.12	$1,080,634.47
Fill rate	80.68%	83.92%
Shortfall penalty	$39,619.42	$34,484.12
Orders diverted	0	2 of 8

The quantum circuit proposes only whether each order should remain at its default DC or move to its selected alternate DC. Inventory allocation, fulfilled cases, penalties, shipping costs, and operational feasibility are determined by the classical MILP recourse model. Because the same recourse model evaluates the default, greedy, exact, and hybrid assignments, their reported results are directly comparable.

1 Why DOM Is a Hard Combinatorial Problem

Every open order carries a default DC, a destination ZIP code, a planning date, and a set of SKU lines. When the default DC can't cover it, a planner has to choose: keep it there and accept the shortfall, or divert the whole order to another DC. The right call weighs fulfilled revenue against the shortfall penalty and the freight cost of shipping from wherever the order actually ends up.
These choices aren't independent. Inventory is tied to a DC/SKU/date combination; dock capacity is shared by every order routed to the same DC on the same day; case-pick and pallet-pick throughput work the same way. Move one order into a DC and every other order competing for the same dock slot or pick capacity feels it. That coupling is what makes this a genuine combinatorial problem rather than 1,109 independent lookups. With M binary decisions there

are 2^M possible assignments, and the eight-order batch used for certification in this report already spans 256 combinations before SKU-level fulfilment is even worked out.
Every default, greedy, exact, and quantum generated assignment is evaluated using the same recourse model. Therefore, all methods are compared under a common business objective and the same operational feasibility checks. The quantum component proposes only the keep or divert decisions; it does not replace or bypass the inventory, fulfilment, capacity, penalty, or shipping calculations.

2 Data and Candidate Construction

2.1 Inputs

Five input files feed the pipeline. The order file provides order SKU lines, the default plant, planning date, destination postal code, ordered quantity, revenue, and cases per pallet information when available. The remaining files provide plant to destination shipping costs, inventory by DC, material, and date, available dock capacity by plant and date, and case pick, pallet pick, and order count information by plant and date. After data cleaning, the working dataset contains $25,193$ order–SKU lines representing $1,109$ distinct orders across eight default DCs.

2.2 Screening

Not every order requires intervention. The screening stage retains only those orders whose default DC cannot fully satisfy demand and for which at least one alternate DC provides a material improvement. An alternate is considered eligible only if it increases the fillable quantity by at least

$$\Delta_o = \max\{0.05, Q_o,\ 100\}$$

cases, where (Q_o) denotes the total demand of order (o). Among the eligible alternates, the model first selects the DC that provides the highest fillable quantity and then uses the net-value proxy as a tie-breaker. Each screened order therefore enters the master problem with two alternatives: remain at its default DC or move to its strongest eligible alternate. This process identifies 100 divert eligible orders.

2.3 Construction of a Contested Evaluation Batch

A meaningful optimization test should include decisions that interact through shared operational resources. Therefore, candidate orders are grouped by their alternate DC and planning date. For each group, the combined dock, case-pick, and pallet-pick loads are evaluated under the scenario in which all competing orders are diverted simultaneously. Orders associated with the strongest shared resource pressure are then prioritized. This procedure produces an eight-order instance containing 95 distinct SKUs, which is used consistently for all methods and results reported in this study.

2.4 Evaluation Settings

The default, greedy, exact, and hybrid methods are evaluated using the same 95% fill-rate threshold and the same 15% value-weighted shortfall penalty rate. Consequently, all methods are compared under a common objective definition and identical operational feasibility conditions. For a plant with no dock-capacity records across the entire data horizon, the dock constraint is treated as non-binding rather than approximated using an unrelated operational measure. Inventory, case-pick, and pallet-pick constraints remain active for all assignments.

3 Deterministic binary formulation

For order o, let d_o^0 be the default DC, d_o^1 the selected alternate, t_o the planning date, and $Q_o = \sum_{s \in S_o} q_{os}$ the total demand across SKUs s in the order.

Master assignment variable

The stay or divert decision is binary:

$$y_o = \begin{cases} 0, & \text{keep order } o \text{ at } d_o^0, \\ 1, & \text{divert order } o \text{ to } d_o^1. \end{cases}$$

For a fixed bitstring y, the assigned DC $d(o;y)$ is known, and the recourse problem below is solved to optimality.

Recourse variables

For every order–SKU line, the recourse model chooses fulfilled cases x_{os}, full pallets m_{os}, loose cases r_{os}, and penalized shortfall u_{os}, with binary indicators v_o (order served) and b_o (order below the fill threshold).

Objective

The objective:

$$\max \; Z(y) = \sum_{o \in O} \sum_{s \in S_o} p_{os}\, x_{os} \; - \; \rho \sum_{o \in O} \sum_{s \in S_o} p_{os}\, u_{os} \; - \; \sum_{o \in O} c_{o,\, d(o;y)}\, v_o,$$

where p_{os} is revenue per fulfilled case, $\rho = 0.15$ is the shortfall penalty rate, and $c_{o,\, d(o;y)}$ is the flat freight cost for whichever DC order o is assigned to under y.

Operational Recourse Constraints

For a fixed master assignment y, let $d(o;y)$ denote the DC selected for order o. The recourse model determines the fulfilled quantity, pallet and loose-case requirements, threshold status, and penalized shortfall while enforcing inventory and operational-capacity limits.

Demand and inventory constraints. The fulfilled quantity of each order–SKU line cannot exceed its demand. In addition, the total quantity of each SKU allocated from a DC on a given planning date cannot exceed the available inventory:

$$0 \le x_{os} \le q_{os}, \qquad \forall o \in O,\ s \in S_o, \tag{1}$$

$$\sum_{\substack{o:\, s \in S_o \\ d(o;y)=d,\ t_o=t}} x_{os} \le I_{dst}, \qquad \forall d \in D,\ s \in S,\ t \in T. \tag{2}$$

Pallet and loose-case decomposition. Each fulfilled quantity is divided exactly into full pallets and loose cases:

$$x_{os} = P_s m_{os} + r_{os}, \qquad \forall o \in O,\ s \in S_o, \tag{3}$$

$$0 \le r_{os} \le P_s - 1, \qquad \forall o \in O,\ s \in S_o, \tag{4}$$

where P_s is the number of cases per pallet, m_{os} is the number of full pallets, and r_{os} is the remaining loose-case quantity.

Operational-capacity constraints. The total loose-case, pallet-pick, and dock usage at each DC and planning date must remain within the corresponding capacities:

$$\sum_{\substack{o,\, s\in S_o \\ d(o;y)=d,\, t_o=t}} r_{os} \le K_{dt}^{\text{case}}, \qquad \forall d \in D,\ t \in T, \tag{5}$$

$$\sum_{\substack{o,\, s\in S_o \\ d(o;y)=d,\, t_o=t}} m_{os} \le K_{dt}^{\text{pallet}}, \qquad \forall d \in D,\ t \in T, \tag{6}$$

$$\sum_{\substack{o:\, d(o;y)=d \\ t_o=t}} v_o \le K_{dt}^{\text{dock}}, \qquad \forall d \in D,\ t \in T. \tag{7}$$

Here, $v_o = 1$ when order o receives a positive fulfilled quantity and therefore uses one dock-processing slot.

Fill-threshold and shortfall constraints. Let

$$F_o = \sum_{s\in S_o} x_{os}, \qquad Q_o = \sum_{s\in S_o} q_{os}, \qquad H_o = \lceil 0.95\, Q_o \rceil .$$

The binary variable b_o identifies whether order o falls below the required fill threshold:

$$F_o \ge H_o(1 - b_o), \qquad \forall o \in O, \tag{8}$$

$$F_o \le H_o - 1 + Q_o(1 - b_o), \qquad \forall o \in O. \tag{9}$$

These constraints enforce $b_o = 0$ when $F_o \ge H_o$ and $b_o = 1$ when $F_o < H_o$.

For each order–SKU line, the penalized shortfall u_{os} is defined by

$$0 \le u_{os} \le q_{os} b_o, \qquad \forall o \in O,\ s \in S_o, \tag{10}$$

$$u_{os} \le q_{os} - x_{os}, \qquad \forall o \in O,\ s \in S_o, \tag{11}$$

$$u_{os} \ge q_{os} - x_{os} - q_{os}(1 - b_o), \qquad \forall o \in O,\ s \in S_o. \tag{12}$$

Thus, $u_{os} = q_{os} - x_{os}$ when the order is below the fill threshold, while $u_{os} = 0$ when the threshold is satisfied. The penalty therefore reflects the actual unfulfilled quantity rather than an independently chosen variable.

The relevant variable domains are

$$x_{os},\ m_{os},\ r_{os},\ u_{os} \in \mathbb{Z}_{\ge 0}, \qquad v_o,\ b_o \in \{0, 1\}.$$

Master recourse design: The PCE layer represents the binary keep-or-divert decisions, while the classical recourse model determines fulfilled cases, pallet and loose-case quantities, and shortfall. This division keeps the quantum representation compact and allows all operational quantities and shared-resource constraints to be evaluated exactly. The hybrid architecture therefore applies correlation-based compression to the assignment layer and uses the MILP for detailed fulfilment and feasibility analysis.

4 Hybrid PCE-MILP Solution Implementation

The proposed workflow consists of five stages: data cleaning and integration, candidate screening and contested-batch construction, training of the three-qubit PCE master, exact MILP recourse evaluation, and final validation with planner-facing output.

The quantum circuit represents only the binary keep-or-divert decisions. For every decoded assignment, the classical recourse model determines fulfilled quantities, pallet and loose-case requirements, shortfall penalties, shipping costs, and compliance with inventory, dock, and throughput constraints. Consequently, all reported business metrics are obtained from the same exact recourse model.

4.1 Degree-two Pauli Correlation Encoding

For degree-two PCE, n_q qubits can support $3\binom{n_q}{2}$ distinct two-body correlators drawn from the XX, YY, and ZZ families. Encoding our eight master routing decisions requires $3\binom{n_q}{2} \geq 8$. Three qubits satisfy this requirement perfectly, providing nine available correlators for our eight variables. We map these eight decisions directly to the Pauli strings IXX, XIX, XXI, IYY, YIY, YYI, IZZ, and ZIZ, leaving the ninth combination unused.

For a given Pauli correlator Π_i, the continuous soft decision used during optimization is calculated from the expectation value of the parameterized quantum state $|\psi(\theta)\rangle$:

$$\tilde{y}_i = \frac{1 - \tanh(\alpha\langle\psi(\theta)|\Pi_i|\psi(\theta)\rangle)}{2}$$

Upon finalizing the quantum parameters, this continuous relaxation is "hardened" into a strict binary assignment by replacing the tanh function with the discrete sign function:

$$y_i = \frac{1 - \mathrm{sign}(\langle\psi(\theta)|\Pi_i|\psi(\theta)\rangle)}{2}$$

Consequently, the final binary decision evaluates to $y_i = 1$ (divert) when the expectation value $\langle\psi(\theta)|\Pi_i|\psi(\theta)\rangle$ is negative, and $y_i = 0$ (keep at default) when it is positive.

The underlying quantum circuit employs a fully entangling `efficient_su2` ansatz with two repetitions, yielding 18 trainable parameters. The sharpening parameter α begins at a baseline of 3.0 to allow smooth, continuous exploration of the solution space. It then grows geometrically by a factor of 1.3 in each outer iteration, progressively forcing the continuous quantum correlations into the strict binary decisions required for the final operational routing.

4.2 Capacity aware training loss

The circuit trains against an Augmented Lagrangian-like variant combined with a hinge loss formulation. Inequality constraints are incorporated into the objective function as a hinge loss, where the Lagrange multipliers are iteratively updated. The loss function is defined as:

$$\mathcal{L}(\theta) = -\frac{V(\tilde{y}(\theta))}{V_{scale}} + \sum_k \left[\lambda_k^{(t)} g_k^+ + \frac{\mu_k}{2}(g_k^+)^2\right]$$

Here, g_k denotes the raw, normalized constraint evaluation (used capacity minus available capacity) prior to applying the maximum function, and $g_k^+ = \max(0, g_k)$ represents the actual hinge loss violation. We refer to this as an "ALM variant" because the formulation assigns an independent penalty parameter (μ_k) to each constraint class (dock, case-pick, pallet-pick) to allow for decoupled tuning, whereas standard ALM typically relies on a single global μ.

For this specific evaluation, these structural penalty weights are uniformly set to $\mu_k = 10.0$. The Lagrange multipliers λ_k, however, are updated independently at the end of each outer iteration t using their respective observed violations:

$$\lambda_k^{(t+1)} = \max\left(0, \lambda_k^{(t)} + \mu_k g_k^+\right)$$

Keeping dock, case-pick, and pallet-pick penalties separate, rather than folding them into one blended term, keeps the training signal honest about which specific constraint is actually binding. Three seeded restarts $(42, 1042, 2042)$, COBYLA capped at 300 evaluations per inner solve. The sharpening parameter α begins at a baseline of 3.0 to allow smooth, continuous exploration of the solution space, growing geometrically by a factor of 1.3 in each outer iteration to progressively harden the quantum correlations into strict binary decisions. The winning candidate comes from restart 2, outer iteration 4, $\alpha = 6.591$, bitstring `10010000`. Later iterations hold the same assignment while the worst constraint violation shrinks toward zero – the noiseless decode already matches the exact optimum, so post-processing makes no further change to it.

4.3 Exact Recourse Evaluation and Local Repair

Each decoded assignment is evaluated using the exact MILP recourse model. The post-processing routine first evaluates the complementary bitstring and then examines all one-bit neighbours of the current assignment. A change is accepted only when it improves the exact recourse objective, and the neighbourhood search continues until no further improvement is found.

Because the certification instance contains only eight binary decisions, this local search adds little computational overhead. It provides a practical safeguard against isolated decoding errors while preserving the same objective function and operational feasibility checks used throughout the study.

5 Baselines and Experimental Design

Four methods run against the same batch and the same recourse model:

- **Default** — every master bit set to 0.
- **Greedy** — orders considered in descending risk order; a flip is kept only if exact recourse shows it improves the objective.
- **Exact** — all 256 bitstrings completed and scored; the certificate against which everything else is measured, not a scalable method on its own.
- **Hybrid PCE–MILP** — the variational circuit proposes a candidate, then exact completion and local repair finish it.

Runtime is measured fairly: the recourse cache is cleared before each method's timing starts, so no method inherits MILP solves performed by another. The noise experiment reuses the trained parameters without retraining — a FakeManilaV2 calibration-based noise model, 20 independent seeds, 4,096 shots per trial, with the same one-bit repair applied to every noisy readout.

6 Results

Method	Z ($)	Fill	Diverts	Penalty ($)	Ship ($)	Runtime (s)	Gap
Default	1,047,384.12	80.68%	0	39,619.42	17,454	0.36	3.08%
Greedy	1,080,634.47	83.92%	2	34,484.12	21,685	3.58	0.00%
Exact (benchmark)	1,080,634.47	83.92%	2	34,484.12	21,685	91.00	0.00%
Hybrid PCE–MILP	1,080,634.47	83.92%	2	34,484.12	21,685	43.73	0.00%

The optimized plan adds $33,250.35 to the objective and a 3.17% lift over default. Fulfilled revenue rises $32,346.05, shortfall penalty drops $5,135.30 (down 12.96%), and freight rises $4,231 because two orders now ship from an alternate DC. That freight increase is real and fully charged against the plan; the objective still comes out ahead. Fill rate climbs 3.24 points, from 80.68% to 83.92%.

Greedy, exact, and hybrid all converge on the same bitstring, `10010000`. That agreement does two things: exhaustive enumeration certifies it as the true optimum, and greedy reaching the identical answer through a much simpler rule shows this particular batch doesn't need anything elaborate to solve well. The value of the hybrid approach here isn't beating greedy on eight orders — it's the qubit-efficient representation and capacity-aware search, delivered at roughly 48% of exhaustive enumeration's runtime.

The two diversions read differently in practice. O4 moves to 99.2% fill and clears its threshold penalty entirely. O1 diverts too, but even at the alternate DC it's still short by a wide margin; the model picks it because diverting reduces the loss relative to staying put, not because it fixes the order. That distinction matters for a planner: it's an escalation, not a resolution.

7 Noise Sensitivity and Classical Repair

Across 20 finite-shot trials, the exact bitstring survives in 9. The other 11 all land on the same single-bit neighbour, 10000000 — no trial is off by more than one bit. Mean bit flips: 0.55. The tightest correlation margin observed is 4.88×10^{-4}, meaning one of the eight decisions sits close enough to its sign boundary that shot noise can flip it.

Before repair, the mean recourse-completed objective is \$1,067,887.54 (σ = \$11,829.56). After the one-bit repair, all 20 trials land back on $\$1,080,634.47$, which shows 100%recovery, zero variance. This is exactly what the hybrid design is for: let the quantum circuit produce a compact, noisy candidate distribution, and let a cheap classical step turn it into something stable enough to hand to a planner.

8 Scalability and Deployment Roadmap

Qubit count for degree-two PCE follows the smallest n_q with $3\binom{n_q}{2} \geq M$: 5 qubits cover 20 master decisions, 7 cover $50, 9$ cover $100, 13$ cover 200. That's register size, not total cost, so the number of correlators to measure and the size of the variational search both grow as the ansatz expands, and statevector simulation stays exponential in physical qubit count regardless of how compact the encoding is.

Exact enumeration (2^M) is only viable at certification scale. It isn't a deployment strategy. The recourse MILP itself scales with the number of positive-demand order–SKU lines and active inventory/capacity keys, a much gentler curve. A practical path to larger volumes is coordinated batching: solve $6-12$ highly contested orders at a time, update residual inventory and capacity, move to the next batch, and run a reconciliation pass to catch anything lost at batch boundaries.

Three natural next steps: let each order carry several alternate DCs and move to a column-selection master formulation; replace the shared 95%/15% parameterisation with order-specific thresholds and penalty schedules; and warm-start the variational search with a greedy solution or a clustering of the order–DC graph before running on larger simulators or real hardware.

9 Scope, Reproducibility, and Future Extensions

This study evaluates the hybrid PCE–MILP framework on a controlled eight-order contested instance. Each focus order is reduced to two master alternatives: its default DC and the strongest eligible alternate identified during screening. This keeps the quantum master compact while preserving the most relevant reassignment decision. The same architecture can be extended to multiple alternate DCs through a larger assignment model.

All methods use the same 95% fill-rate threshold, 15% value-weighted shortfall penalty, objective function, and operational feasibility checks. Throughput data are aggregated by plant and planning date for case-pick and pallet-pick constraints. Dock capacity is enforced wherever records are available; plants without a complete dock feed are handled using a clearly defined

non-binding rule rather than an unrelated proxy.
Exact enumeration evaluates all $2^8 = 256$ assignments and therefore certifies the optimum for the reported instance. For larger problems, the framework can scale through candidate reduction, coordinated batching, or decomposition, while retaining exact MILP evaluation of each proposed assignment.

Reproducibility

The implementation uses Python, pandas, NumPy, PuLP/CBC, SciPy COBYLA, Qiskit, and Qiskit Aer. The global seed is 42, with variational restarts initialized using seeds $42, 1042$, and 2042. The PCE model uses three qubits, degree-two Pauli correlations, two ansatz repetitions, and three restarts. Noise evaluation uses 20 independent FakeManilaV2 trials with $4,096$ shots per trial.
For fair runtime comparison, the recourse cache is cleared before timing each method. The saved outputs include the method comparison, PCE mapping, optimization history, post-processing trace, noise trials, scaling analysis, and anonymized planner recommendations, allowing the reported objective, assignment, optimality gap, and repair performance to be verified.